Name: Mark Scheme.

Teacher: ____________________

Class: ____________ Mark: /25

Baldivis
Secondary College

YEAR 8
SEMESTER 1 2015
Algebra Test
Resources Allowed

Question 1 **[3]**

In the expression $7x^2 + 5x - 2$

(a) How many terms are there?

3 ✓

(b) What is the coefficient of x?

5 ✓

(c) What is the constant term?

-2 ✓

Question 2 **[2]**

Philip thought of a number (n), he then doubles it and adds 10.

(a) Write an expression for this situation

$2n+10$ ✓

(b) Find the value of this expression when n is equal to 12.

34. ✓

Question 3 **[7]**

Simplify the following:

(a) $2x + 8x$ $= 10x$ ✓

(b) $5x + y - x - 4y$ $4x - 3y$. ✓

(c) $2m \times 8n$ $16mn$. ✓

(d) $5g \times 7g$ $35g^2$ ✓

END OF TEST

(e) $\frac{\cancel{5}tu}{\cancel{25}uv}$ 5

$\frac{t}{5v}$ ✓

(f) $\frac{6a^2b}{2ab^2}$

$\frac{3a}{b}$ ✓ ✓

Question 4 **[7]**

Expand and simplify:

(a) $5(x+3)$

$5x+15$ ✓

(b) $-2(y-7)$

$-2y+14$ ✓

(c) $2x(4x-7y+3)$

$8x^2-14xy+6x$ ✓

(d) $3(a+2)+3a$

$3a+6+3a = 6a+6$ ✓ ✓

(e) $5(x+3)+2(x+6)$

$5x+15+2x+12$ ✓ $= 7x+27$ ✓

Question 5 **[6]**

Factorise:

(a) $5x+15$ $5(x+3)$ ✓

(b) $2x-6$ $2(x-3)$ ✓

(c) $10y+4z$ $2(5y+2z)$ ✓

(d) x^2+6x $x(x+6)$ ✓

(e) $4x^2+2x+10$

$2(x^2+x+5)$ ✓ ✓

END OF TEST